[INCH–POUND]
A–A–59590/9B
21 June 2010
SUPERSEDING
A-A-59590/9A
12 August 2005

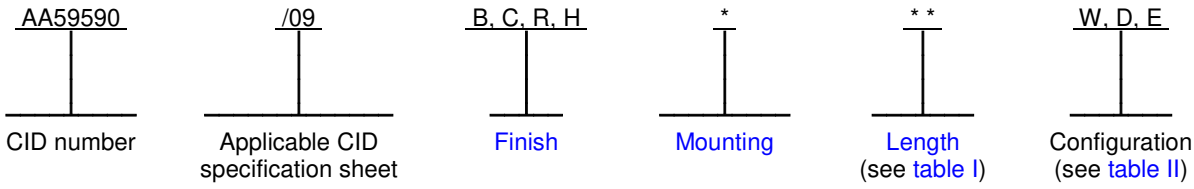
COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .220 X .220 INCH BODY SIZE, WITH SCREW RETENTION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the issue in effect of A–A–59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).



Example: AA59590/09BV28E is the PIN for a hard black anodize finished, 2.8 inch (71.1 mm) long card holder with a retained actuating screw. The card holder also features two tapped mounting holes for use with 2–56 UNC 2B fasteners and a screw self-lock element for added resistance to loosening.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G-rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces. The screw retention feature is to prevent unintentional disassembly of the card holder.

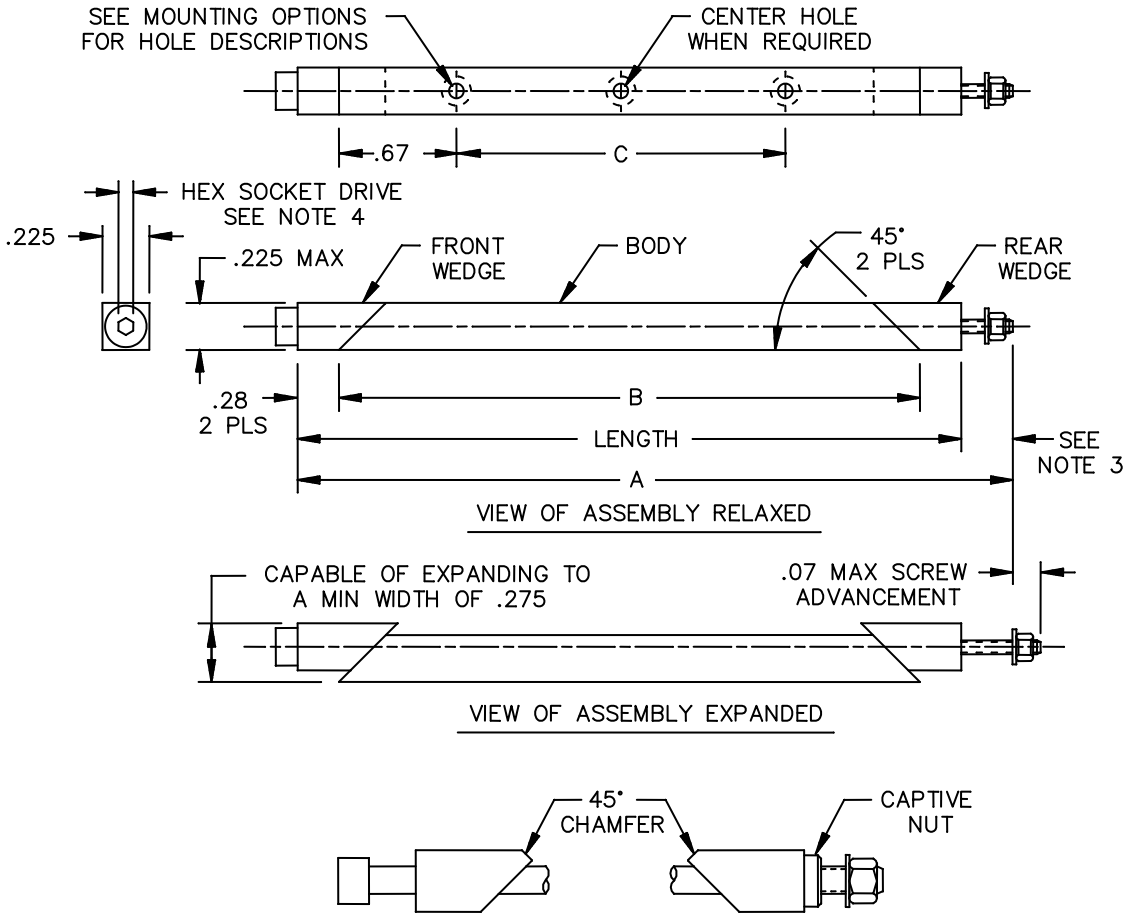
Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, 4, in table 1, and A–A–59590.

Actuating screw hex drive socket. The across flats dimension for the hex drive socket shall be as follows: .094 inch (2.38 mm) for mounting options "J", "N", "T", "R", "V", and "S" and 2.5 mm (0.98 inch) for mounting options "G" and "M".

Material. Unless otherwise specified herein, the card holder materials shall be as specified in A–A–59590.

Screw retention (locknut). The screw retention feature intended to prevent unintentional disassembly of the card holder shall be as specified in figure 2. The locknut shall be fabricated of a corrosion resistant material.

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Inches	mm	Inches	mm	Inches	mm
.07	1.8	.225	5.72	.28	7.1
.220	5.59	.275	6.99	.67	17.0

NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.5 mm) for two place decimals and ±.010 (0.25 mm) for three place decimals.
3. Dimension shall be .20 (5.1 mm) maximum on card holders of the base configuration or configuration "E" and .30 (7.6 mm) maximum on card holders of configuration "W" or "D".
4. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm) for mounting options "J", "N", "T", "R", "V", and "S". The across flats dimension for hex drive socket shall be a metric dimension of 2.5 mm (0.98 inch) across flats for mounting options "G" and "M".
5. An alternate design for the front and rear wedge is used on CAGE 18915 parts. The overall length of the assembly will be reduced by approximately .030 inch (0.76 mm).

FIGURE 1. Relaxed and expanded dimensions.

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FIGURE 2. Screw retaining self-locking nut (corrosion resistant) details.

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 6 inch–pounds (0.68 N–m) for assemblies of the base configuration or configuration "W" and 8 inch–pounds (0.90 N–m) for assemblies of configurations "E" or "D".

Cold plate dimensions. Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .250 inch (6.35 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

Finish. The finish materials shall be as specified in A–A–59590. The finishes available for this CID specification sheet are as follows: "B" (black anodize), "C" (gold chemical film), "R" (clear chemical film), or "H" (hard black anodize).

Mounting. The mounting designators shall be as specified in A–A–59590. The mounting options available for this CID specification sheet are as follows: "J" (rivet mounting holes), "N" (no mounting holes), "T" or "R" (tapped 0–80 UNF 2B holes), "V" or "S" (tapped 2–56 UNC 2B holes), or "G" or "M" (tapped M2.5 x.45 metric holes). See table I and figure 1 for mounting hole spacing requirements. Card holders using mounting option "J" may be shipped unassembled.

Mounting holes and locations (when required). Two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole shall be centered on the mounting body. See figure 1 and table I for mounting hole locations and spacing requirements.

Rivet mounting holes. The holes used for rivet mounting shall be .066 to .073 inch (1.68 to 1.85 mm) diameter through holes, countersunk 100 degrees by .140 inch (3.56 mm) diameter with an access/clearance counterbore hole of .156 inch (3.96 mm) diameter by .160 to .175 inch (4.06 to 4.45 mm) deep.

Rivets. This card holder uses rivet style A as specified in A–A–59590 when rivet mounting is used.

Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A–A–59590 and the available lengths for this CID specification sheet are listed in table I.

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TABLE I. Additional assembly dimensions (see figure 1). 1/

PIN length designator	Dimension "LENGTH" 2/ ±.02 (0.5)	Dimension "A" (maximum)	Dimension "B" ±.03 (0.8)	Dimension "C" ±.005 (0.13)
18	1.8 (45.7)	2.20 (55.9)	1.24 (31.5)	.75 (19.1)
23	2.3 (58.4)	2.70 (68.6)	1.74 (44.2)	.40 (10.2)
28	2.8 (71.1)	3.20 (81.3)	2.24 (56.8)	.90 (22.9)
30	3.0 (76.2)	3.40 (86.4)	2.44 (62.0)	1.10 (27.9)
33	3.3 (83.8)	3.70 (94.0)	2.74 (69.6)	1.40 (35.6)
35	3.5 (88.9)	3.90 (99.1)	2.94 (74.7)	1.60 (40.6)
38	3.8 (96.5)	4.20 (106.7)	3.24 (82.3)	1.90 (48.3)
43	4.3 (109.2)	4.70 (119.4)	3.74 (95.0)	2.40 (61.0)
48	4.8 (121.9)	5.20 (132.1)	4.24 (107.7)	2.90 (73.7)
53	5.3 (134.6)	5.70 (144.8)	4.74 (120.4)	3.40 (86.4)
58	5.8 (147.3)	6.20 (157.5)	5.24 (133.1)	3.90 (99.1)
63	6.3 (160.0)	6.70 (170.2)	5.74 (145.8)	4.40 (111.8)
68	6.8 (172.7)	7.20 (182.9)	6.24 (158.5)	4.90 (124.5)
73	7.3 (185.4)	7.70 (195.6)	6.74 (171.2)	5.40 (137.2)
78	7.8 (198.1)	8.20 (208.3)	7.24 (183.9)	5.90 (149.9)
83	8.3 (210.8)	8.70 (221.0)	7.74 (196.6)	6.40 (162.6)
88	8.8 (223.5)	9.20 (233.7)	8.24 (209.3)	6.90 (175.3)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.
2/ The overall length of the assembly supplied by CAGE 18915 will be reduced by approximately .030 inch (0.76 mm) due to an alternate design for the front and rear wedge.

Configuration. The configuration of a card holder shall be as specified in table II. The details of a particular configuration consist of those on figures 1, 2, 3, and 4. Card holders not requiring the options described by table II shall leave the configuration position in the PIN blank.

TABLE II. Configurations.

Configuration	Applicable figures	Hardware options
---	1 and 2	No added options
W	1, 2 and 3	Lockwasher and flat washer
E	1, 2 and 4	Screw self-locking element
D	1, 2, 3 and 4	Lockwasher, flat washer and screw self-locking element

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Lockwasher and flat washer (see figure 3). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer shall include a suffix "W" in the PIN (see classification and notes).

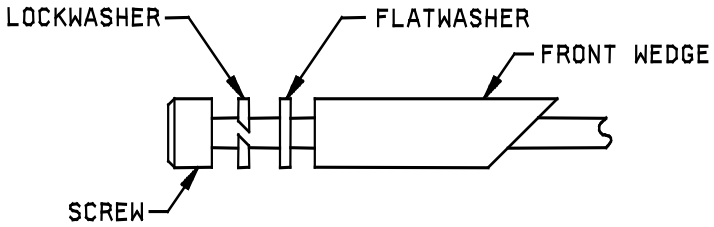


FIGURE 3. Lockwasher and flat washer details.

Screw self-locking element (see figure 4). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix "E" in the PIN (see classification and notes).



FIGURE 4. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element configuration shall include a suffix "D" in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Ordering data. Ordering data is as specified in A–A–59590.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

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Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	Birtcher – A Division of Pentair Electronic Packaging 7238 Trade Street San Diego, CA 92121–3410	Telephone: (858) 740–2400 Toll Free: (800) 854–7086 Facsimile: (858) 679–4555 (URL): www.birtcherproducts.com
52094	Calmark – A Division of Pentair Electronic Packaging 7238 Trade Street San Diego, CA 92121–3410	Telephone: (858) 740–2400 Toll Free: (800) 854–7086 Facsimile: (858) 740–2430 Electronic mail: sales@calmark.com URL: www.calmark.com
5BG68	Card Locks Unlimited, Inc. 2310 E. Orangethorpe Avenue Anaheim, CA 92806–1231	Telephone: (714) 738–6194 Facsimile: (714) 446–0119 E-mail: sales@clumfg.com URL: www.clumfg.com

Part number supersession data. This CID specification sheet PINs supersede the following manufacturer's part numbers as shown in table III. The CID PINs listed in table II are only for length designator "28". See table IV for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE III. Commercial part number supersession data.

PIN designator AA59590/09	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094	Vendor similar designator or type part number 1/ CAGE 5BG68
BJ28	42–6–B–LN	A230CR2.80HR	3230BA–2.80HR
BN28	42–6–B–LN–A	A230CR2.80R	3230BA–2.80R
BT28	42–6–B–LN–T	A230CR2.80T0R	3230BA–2.80T0R
BV28	42–6–B–LN–S	A230CR2.80T2R	3230BA–2.80T2R
BG28	42–6–B–LN–M	MA230CR2.80TM2.5R	3230MBA–2.80TM2.5R
BJ28E	42–6–B–LN–L	A230CR2.80HLR	3230BA–2.80HLR
BN28E	42–6–B–LN–A–L	A230CR2.80LR	3230BA–2.80LR
BT28E	42–6–B–LN–T–L	A230CR2.80T0LR	3230BA–2.80T0LR
BV28E	42–6–B–LN–S–L	A230CR2.80T2LR	3230BA–2.80T2LR
BG28E	42–6–B–LN–M–L	MA230CR2.80TM2.5LR	3230BA–2.80TM2.5LR
BJ28W	42–6–B–LF–LN	WA230CR2.80HR	3230WBA–2.80HR
BN28W	42–6–B–LF–LN–A	WA230CR2.80R	3230WBA–2.80R
BT28W	42–6–B–LF–LN–T	WA230CR2.80T0R	3230WBA–2.80T0R
BV28W	42–6–B–LF–LN–S	WA230CR2.80T2R	3230WBA–2.80T2R
BG28W	42–6–B–LF–LN–M	MWA230CR2.80TM2.5R	3230MWBA–2.80TM2.5R

See footnote at end of table.

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TABLE III. Commercial part number supersession data – Continued.

PIN designator AA59590/09	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094	Vendor similar designator or type part number 1/ CAGE 5BG68
BJ28D	42–6–B–LF–LN–L	WA230CR2.80HLR	3230WBA–2.80HLR
BN28D	42–6–B–LF–LNAL	WA230CR2.80LR	3230WBA–2.80LR
BT28D	42–6–B–LF–LNTL	WA230CR2.80T0LR	3230WBA–2.80T0LR
BV28D	42–6–B–LF–LNSL	WA230CR2.80T2LR	3230WBA–2.80T2LR
BG28D	42–6–B–LF–LNML	MWA230CR2.80TM2.5LR	3230MWBA–2.80TM2.5LR
CJ28	42–6–LN	230CR2.80HR	3230CG–2.80HR
CN28	42–6–LN–A	230CR2.80R	3230CG–2.80R
CT28	42–6–LN–T	230CR2.80T0R	3230CG–2.80T0R
CV28	42–6–LN–S	230CR2.80T2R	3230CG–2.80T2R
CG28	42–6–LN–M	M230CR2.80TM2.5R	3230MCG–2.80TM2.5R
CJ28E	42–6–LN–L	230CR2.80HLR	3230CG–2.80HLR
CN28E	42–6–LN–A–L	230CR2.80LR	3230CG–2.80LR
CT28E	42–6–LN–T–L	230CR2.80T0LR	3230CG–2.80T0LR
CV28E	42–6–LN–S–L	230CR2.80T2LR	3230CG–2.80T2LR
CG28E	42–6–LN–M–L	M230CR2.80TM2.5LR	3230MCG–2.80TM2.5LR
CJ28W	42–6–LF–LN	W230CR2.80HR	3230WCG–2.80HR
CN28W	42–6–LF–LN–A	W230CR2.80R	3230WCG–2.80R
CT28W	42–6–LF–LN–T	W230CR2.80T0R	3230WCG–2.80T0R
CV28W	42–6–LF–LN–S	W230CR2.80T2R	3230WCG–2.80T2R
CG28W	42–6–LF–LN–M	MW230CR2.80TM2.5R	3230MWCG–2.80TM2.5R
CJ28D	42–6–LF–LN–L	W230CR2.80HLR	3230WCG–2.80HLR
CN28D	42–6–LF–LN–A–L	W230CR2.80LR	3230WCG–2.80LR
CT28D	42–6–LF–LN–T–L	W230CR2.80T0LR	3230WCG–2.80T0LR
CV28D	42–6–LF–LN–S–L	W230CR2.80T2LR	3230WCG–2.80T2LR
CG28D	42–6–LF–LN–M–L	MW230CR2.80TM2.5LR	3230MWCG–2.80TM2.5LR
RJ28	42–6–CC–LN	R230CR2.80HR	3230CC–2.80HR
RN28	42–6–CC–LN–A	R230CR2.80R	3230CC–2.80R
RT28	42–6–CC–LN–T	R230CR2.80T0R	3230CC–2.80T0R
RV28	42–6–CC–LN–S	R230CR2.80T2R	3230CC–2.80T2R
RG28	42–6–CC–LN–M	RM230CR2.80TM2.5R	3230MCC–2.80TM2.5R

See footnote at end of table.

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TABLE III. Commercial part number supersession data – Continued.

PIN designator AA59590/09	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094	Vendor similar designator or type part number 1/ CAGE 5BG68
RJ28E	42–6–CC–LN–L	R230CR2.80HLR	3230CC–2.80HLR
RN28E	42–6–CC–LN–A–L	R230CR2.80LR	3230CC–2.80LR
RT28E	42–6–CC–LN–T–L	R230CR2.80T0LR	3230CC–2.80T0LR
RV28E	42–6–CC–LN–S–L	R230CR2.80T2LR	3230CC–2.80T2LR
RG28E	42–6–CC–LN–M–L	RM230CR2.80TM2.5LR	3230MCC–2.80TM2.5LR
RJ28W	42–6–CC–LN–LF	WR230CR2.80HR	3230WCC–2.80HR
RN28W	42–6–CC–LF–LNA	WR230CR2.80R	3230WCC–2.80R
RT28W	42–6–CC–LF–LNT	WR230CR2.80T0R	3230WCC–2.80T0R
RV28W	42–6–CC–LF–LNS	WR230CR2.80T2R	3230WCC–2.80T2R
RG28W	42–6–CC–LF–LNM	MWR230CR2.80TM2.5R	3230MWCC–2.80TM2.5R
RJ28D	42–6–CC–LF–LNL	WR230CR2.80HLR	3230WCC–2.80HLR
RN28D	42–6–CC–LFLNAL	WR230CR2.80LR	3230WCC–2.80LR
RT28D	42–6–CC–LFLNTL	WR230CR2.80T0LR	3230WCC–2.80T0LR
RV28D	42–6–CC–LFLNSL	WR230CR2.80T2LR	3230WCC–2.80T2LR
RG28D	42–6–CC–LFLNML	MWR230CR2.80TM2.5LR	3230MWCC–2.80TM2.5LR
HJ28	42–6–B3–LN	HA230CR2.80HR	3230HA–2.80HR
HN28	42–6–B3–LN–A	HA230CR2.80R	3230HA–2.80R
HT28	42–6–B3–LN–T	HA230CR2.80T0R	3230HA–2.80T0R
HV28	42–6–B3–LN–S	HA230CR2.80T2R	3230HA–2.80T2R
HG28	42–6–B3–LN–M	MHA230CR2.80TM2.5R	3230MHA–2.80TM2.5R
HJ28E	42–6–B3–LN–L	HA230CR2.80HLR	3230HA–2.80HLR
HN28E	42–6–B3–LN–A–L	HA230CR2.80LR	3230HA–2.80LR
HT28E	42–6–B3–LN–T–L	HA230CR2.80T0LR	3230HA–2.80T0LR
HV28E	42–6–B3–LN–S–L	HA230CR2.80T2LR	3230HA–2.80T2LR
HG28E	42–6–B3–LN–M–L	MHA230CR2.80TM2.5LR	3230MHA–2.80TM2.5LR
HJ28W	42–6–B3–LN–LF	WHA230CR2.80HR	3230WHA–2.80HR
HN28W	42–6–B3–LF–LNA	WHA230CR2.80R	3230WHA–2.80R
HT28W	42–6–B3–LF–LNT	WHA230CR2.80T0R	3230WHA–2.80T0R
HV28W	42–6–B3–LF–LNS	WHA230CR2.80T2R	3230WHA–2.80T2R
HG28W	42–6–B3–LF–LNM	MWHA230CR2.80TM2.5R	3230MWHA–2.80TM2.5R

See footnote at end of table.

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TABLE III. Commercial part number supersession data – Continued.

PIN designator AA59590/09	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094	Vendor similar designator or type part number 1/ CAGE 5BG68
HJ28D	42–6–B3–LF–LNL	WHA230CR2.80HLR	3230WHA–2.80HLR
HN28D	42–6–B3–LFLNAL	WHA230CR2.80LR	3230WHA–2.80LR
HT28D	42–6–B3–LFLNTL	WHA230CR2.80T0LR	3230WHA–2.80T0LR
HV28D	42–6–B3–LFLNSL	WHA230CR2.80T2LR	3230WHA–2.80T2LR
HG28D	42–6–B3–LFLNML	MWHA230CR2.80TM2.5LR	3230MWHA–2.80TM2.5LR

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

TABLE IV. Example of PIN with available length designators.

PIN designator AA59590/09	Vendor similar designator or type part number 1/ 2/ CAGE 18915	Vendor similar designator or type part number 1/ 2/ CAGE 52094	Vendor similar designator or type part number 1/ 2/ CAGE 5BG68
BV18E	42–4–B–LN–S–L		3230BA–1.80T2LR
BV23E	42–5–B–LN–S–L	A230CR2.30T2LR	3230BA–2.30T2LR
BV28E	42–6–B–LN–S–L	A230CR2.80T2LR	3230BA–2.80T2LR
BV30E	3/	A230CR3.00T2LR	3230BA–3.00T2LR
BV33E	42–7–B–LN–S–L	A230CR3.30T2LR	3230BA–3.30T2LR
BV35E	3/	A230CR3.50T2LR	3230BA–3.50T2LR
BV38E	42–8–B–LN–S–L	A230CR3.80T2LR	3230BA–3.80T2LR
BV43E	42–9–B–LN–S–L	A230CR4.30T2LR	3230BA–4.30T2LR
BV48E	42–10–B–LN–S–L	A230CR4.80T2LR	3230BA–4.80T2LR
BS53E	42–11–B–LN–S–L	A230CR5.30ET2LR	3230BA–5.30ET2LR
BS58E	42–12–B–LN–S–L	A230CR5.80ET2LR	3230BA–5.80ET2LR
BS63E	42–13–B–LN–S–L	A230CR6.30ET2LR	3230BA–6.30ET2LR
BS68E	42–14–B–LN–S–L	A230CR6.80ET2LR	3230BA–6.80ET2LR
BS73E	42–15–B–LN–S–L	A230CR7.30ET2LR	3230BA–7.30ET2LR
BS78E	42–16–B–LN–S–L	A230CR7.80ET2LR	3230BA–7.80ET2LR
BS83E	42–17–B–LN–S–L	A230CR8.30ET2LR	3230BA–8.30ET2LR
BS88E	42–18–B–LN–S–L	A230CR8.80ET2LR	3230BA–8.80ET2LR

See footnotes at end of table.

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TABLE IV. Example of PIN with available length designators – Continued.

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.
3/ Consult manufacturer for part number.

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Review Activities:
Army – MI
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FAS

Preparing Activity:
DLA – CC

Project 5998–2009–063

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil/.

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